IoT (Internet of Things)

UNIT-I: Introduction to IoT

1. Define IoT. Explain the characteristics and benefits of IoT.
2. Describe the evolution and history of IoT.
3. Explain the basic architecture of IoT.
4. Discuss various applications of IoT in real life.
5. Write a short note on the role of sensors and actuators in IoT.

UNIT-II: IoT Architecture & Design

1. Explain the five-layer architecture of IoT in detail.
2. Describe the enabling technologies of IoT.
3. What is IoT communication model? Explain with a diagram.
4. Explain different communication protocols used in IoT.
5. Compare M2M and IoT.

UNIT-III: IoT Communication Technologies

1. Explain the role of RFID, NFC, Bluetooth, and ZigBee in IoT.
2. Describe LPWAN technologies (e.g., LoRa, SigFox).
3. Discuss the significance of IPv6 in IoT.
4. Explain the concept of MQTT and CoAP protocols.
5. Compare Wi-Fi and Cellular communication in IoT.

UNIT-IV: IoT Platforms and Cloud Integration

1. What is an IoT platform? Explain its key features.
2. Discuss cloud computing integration in IoT systems.
3. Explain the role of Fog and Edge computing in IoT.
4. Describe any one open-source IoT platform.
5. What is the significance of data analytics in IoT?

UNIT-V: IoT Applications & Security

1. Explain IoT applications in smart cities, smart homes, and healthcare.
2. What are the major security challenges in IoT?
3. Discuss privacy issues in IoT environments.
4. Explain data security solutions in IoT.
5. What are the ethical considerations in Iot?